Practice workbook · Explain how biceps, triceps and tendons move the arm

Antagonistic muscle action and tendons moving the forearm at the elbow.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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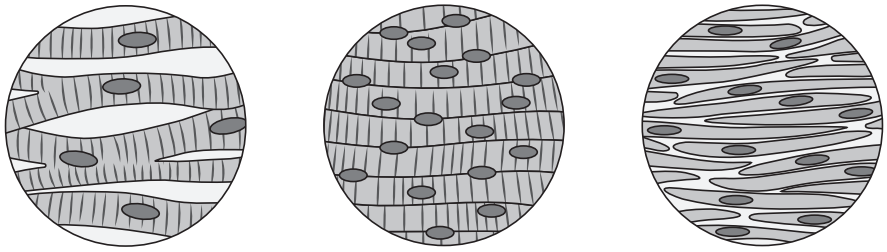
Type ID: HBI-ANTAGONISTIC-MUSCLE-MOVEMENT

1. Question 6 · part c · 4 marks

November 2024 Paper 1

Worked-example retry · Attempt part c only.

6 (a) The diagram shows three types of muscle tissue.



cardiac muscle | voluntary muscle | involuntary muscle

(Source: © udaix/Shutterstock)

(i) Describe the structure of each type of muscle tissue.

(6)

cardiac

voluntary

involuntary

(ii) Give one structure of the human body where involuntary muscle tissue is found.

(1)

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1. Question 6 · part c · 4 marks

November 2024 Paper 1

Worked-example retry · Attempt part c only.

(b) Magnetic resonance imaging (MRI) can be used to measure the percentage muscle mass of an individual.

In one year, 20 female athletes had their percentage muscle mass measured using MRI.

These are the results.

74%	75%	70%	82%	74%	74%	78%	82%	70%	82%
78%	74%	82%	78%	82%	75%	84%	72%	82%	72%

(i) Calculate the mean percentage muscle mass.

(2)

mean percentage muscle mass = ...%

(ii) Draw a frequency table to show the number of each percentage muscle mass recorded for the female athletes.

(3)

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(iii) Determine the mode of these percentage muscle masses.

(1)

mode = ...

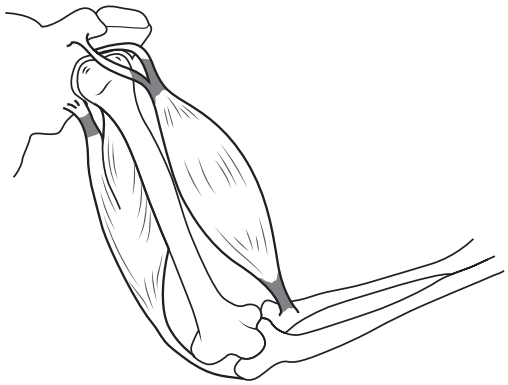
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1. Question 6 · part c · 4 marks

November 2024 Paper 1

Worked-example retry · Attempt part c only.

(c) The diagram shows two muscles in the arm.



Explain how these muscles help to raise the forearm.

(4)

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(Total for Question 6 = 17 marks)

1. Question 6 · part c · 4 marks · Working and answer

Working and answer

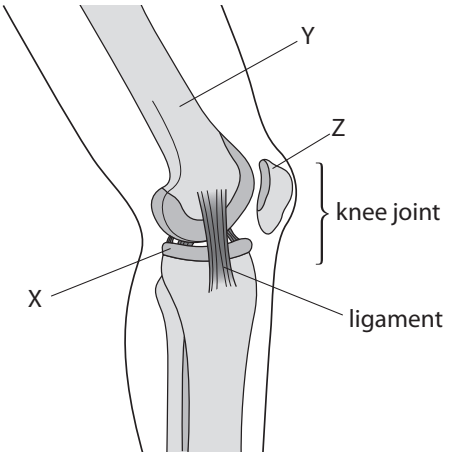
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2. Question 1 · part c · 2 marks

June 2019 Paper 1

Further practice · Attempt part c only.

1 (a) The diagram shows structures in part of a human leg.



(i) The box lists structures in the human leg.

cartilage	femur	kneecap	muscle
fibula	tendon	tibia	

Use words from the box to name structures X, Y, and Z.

(3)

X .....

Y .....

Z .....

(ii) What is the function of ligaments in the knee joint?

(1)

- ☒ **A** ligaments attach muscles to the leg bones
- ☒ **B** ligaments hold the leg bones together
- ☒ **C** ligaments move the leg bones
- ☒ **D** ligaments stop the leg bones rubbing together

(iii) The type of joint found in the knee and the elbow is the same.

Give the name of this type of joint.

(1)

.....

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2. Question 1 · part c · 2 marks

June 2019 Paper 1

Further practice · Attempt part c only.

(b) Aerobic respiration releases energy for muscles to contract.

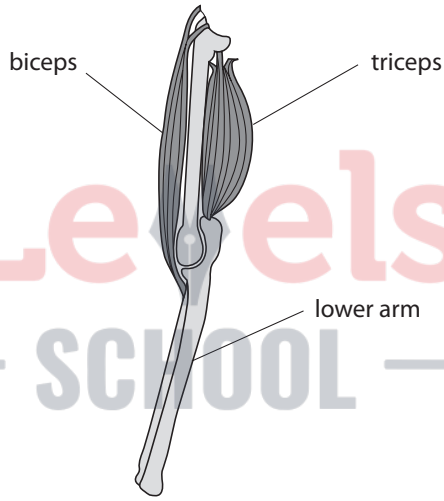
Complete the word equation for aerobic respiration.

(2)

glucose + ... → ..... + water

(c) Diagram 1 shows a model of the human arm, with the lower arm extended.

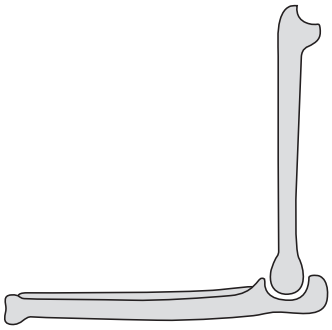
Diagram 1



Complete diagram 2 to show how the muscles change when the lower arm is raised.

(2)

Diagram 2



(Total for Question 1 = 9 marks)

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2. Question 1 · part c · 2 marks · Working and answer

Working and answer

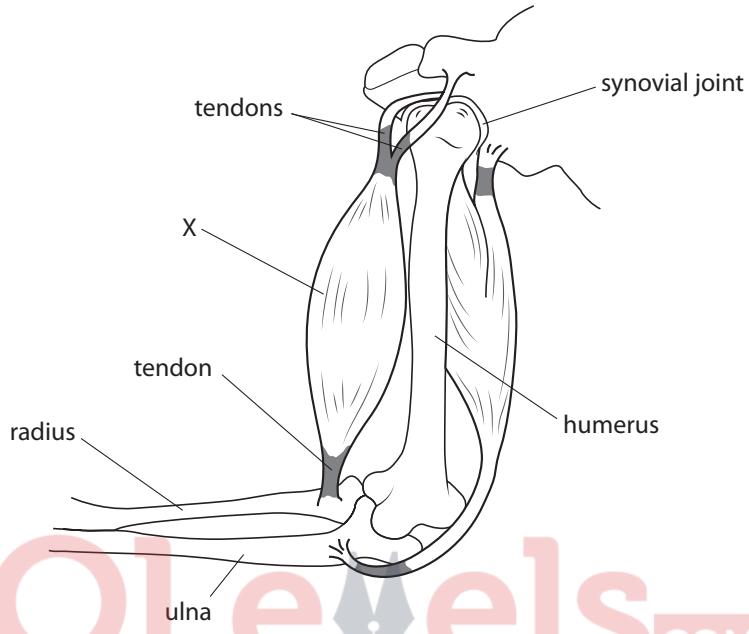
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3. Question 7 · part a.ii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

7 The diagram shows the bones of a human arm and the muscles that move the lower arm.



(a) (i) Muscle X needs energy to contract.

Name the process that supplies this energy.

(1)

.....

(ii) Muscle X contracts when the lower arm is raised further.

Describe the changes in shape of muscle X when the lower arm is raised further.

(2)

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.....

.....

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3. Question 7 · part a.ii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

(iii) Explain how the contraction of muscle X causes the lower arm to be raised.

(3)

(b) Draw a labelled diagram to show the internal structure of the synovial joint.

(6)

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(Total for Question 7 = 12 marks)

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3. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

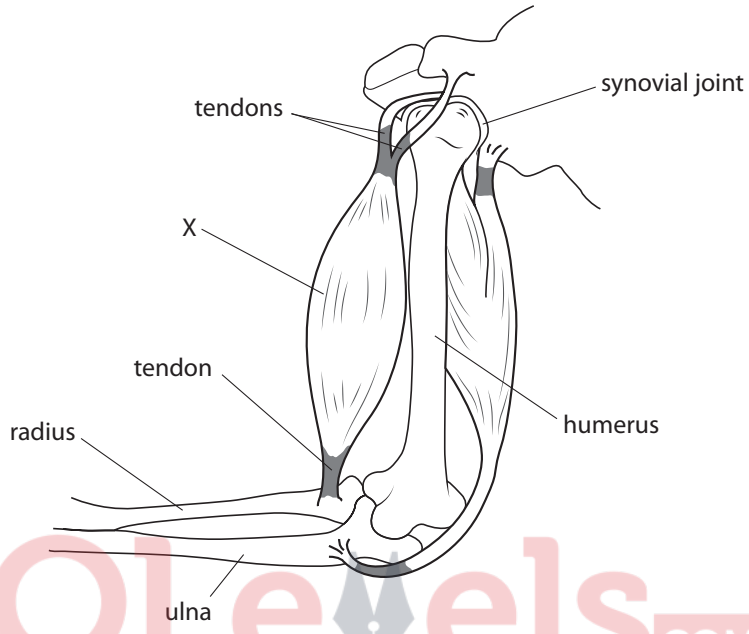
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4. Question 7 · part a.iii · 3 marks

November 2020 Paper 1

Further practice · Attempt part a.iii only.

7 The diagram shows the bones of a human arm and the muscles that move the lower arm.



(a) (i) Muscle X needs energy to contract.

Name the process that supplies this energy.

(1)

.....

(ii) Muscle X contracts when the lower arm is raised further.

Describe the changes in shape of muscle X when the lower arm is raised further.

(2)

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4. Question 7 · part a.iii · 3 marks

November 2020 Paper 1

Further practice · Attempt part a.iii only.

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(iii) Explain how the contraction of muscle X causes the lower arm to be raised.

(3)

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.....

(b) Draw a labelled diagram to show the internal structure of the synovial joint.

(6)

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(Total for Question 7 = 12 marks)

4. Question 7 · part a.iii · 3 marks · Working and answer

Working and answer

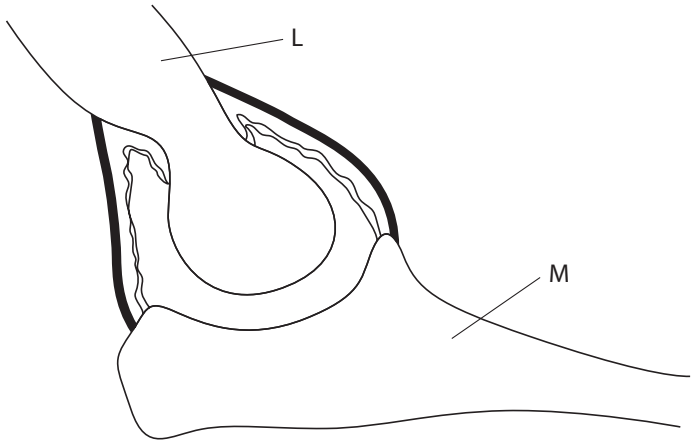
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5. Question 6 · part d · 4 marks

November 2021 Paper 2

Further practice · Attempt part d only.

6 The diagram shows some of the structures of the elbow joint.



(a) State what is meant by the term **joint**.

(2)

.....

.....

.....

.....

(b) (i) Give the name of bone L and the name of bone M.

(2)

L .....

M .....

(ii) Cartilage has not been shown on the diagram.

Draw the missing cartilage on the diagram.

(2)

20

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5. Question 6 · part d · 4 marks

November 2021 Paper 2

Further practice · Attempt part d only.

(c) Explain the effects on a person if the cartilage in this joint is missing or damaged.

(2)

(d) Tendons are important in the functioning of a joint.

Explain the importance of tendons in the movement of bone M.

(4)

(Total for Question 6 = 12 marks)

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5. Question 6 · part d · 4 marks · Working and answer

Working and answer

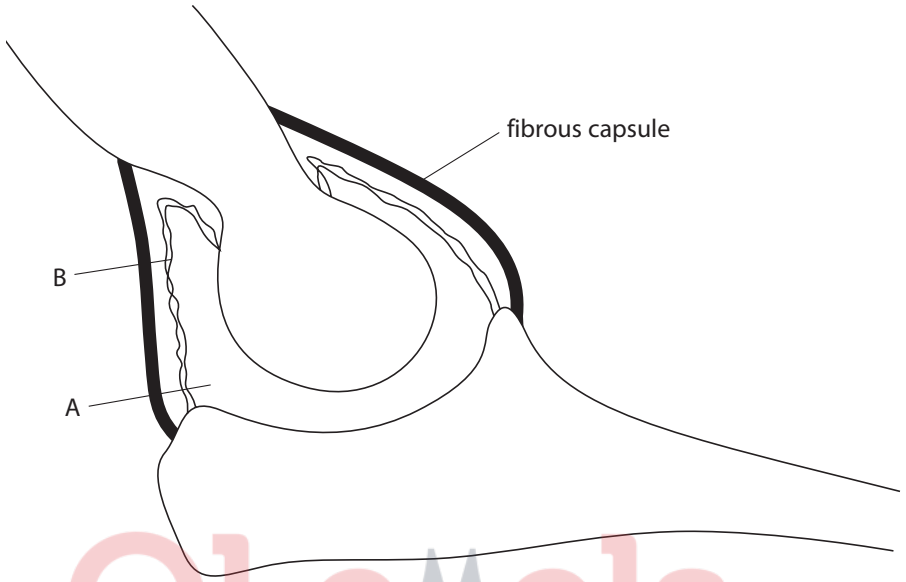
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6. Question 3 · part c · 5 marks

January 2022 Paper 1

Further practice · Attempt part c only.

3 The diagram shows part of an elbow joint without any cartilage.



(a) (i) Identify part A and part B. **(2)**

part A .....

part B .....

(ii) Describe the function of part A. **(2)**

.....

.....

.....

.....

(b) (i) Complete the diagram by drawing the cartilage. **(2)**

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P 6 7 0 5 9 A 0 8 2 4

6. Question 3 · part c · 5 marks

January 2022 Paper 1

Further practice · Attempt part c only.

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(ii) Cartilage often breaks down as people become older.

Explain the effects this has on the working of the joint.

(2)

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(c) Explain the role of tendons in the movement of the lower arm.

(5)

(Total for Question 3 = 13 marks)

6. Question 3 · part c · 5 marks · Working and answer

Working and answer

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7. Question 10 · part b · 3 marks

June 2022 Paper 1R

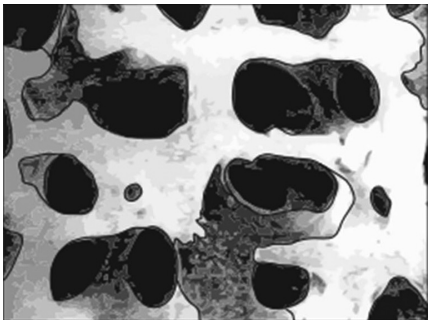
Further practice · Attempt part b only.

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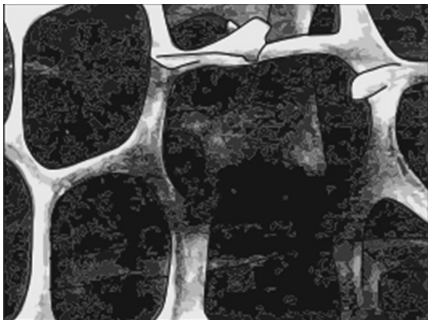
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10 (a) The photographs show the structure of healthy bone and bone that is damaged by osteoporosis.



healthy bone



bone damaged by osteoporosis

Describe how osteoporosis might affect an individual.

(3)

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(b) Explain how voluntary muscles and bones bring about movement in the elbow joint.

(3)

(Total for Question 10 = 6 marks)

7. Question 10 · part b · 3 marks · Working and answer

Working and answer

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8. Question 1 · part whole · 6 marks

June 2025 Paper 1

Further practice · Attempt part whole only.

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1 The sentences below are about muscles and movement of the arm.

Complete the sentences by using words from the box.

(6)

antagonistic	ball and socket	biceps	cartilage
complementary	fixed	hinge	ligament
tendons	triceps		

To cause the movement of a joint, muscles work in pairs, known as

..... pairs of muscles.

The elbow is a type of synovial joint known as a ..... joint.

This joint allows the forearm to be raised when the .....

muscle contracts, and to be straightened when the ..... muscle contracts.

Both of these muscles are connected to bones by ..... .

Smooth movement of the joint is possible because of the presence of

..... covering the ends of the bones.

(Total for Question 1 = 6 marks)

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8. Question 1 · part whole · 6 marks · Working and answer

Working and answer

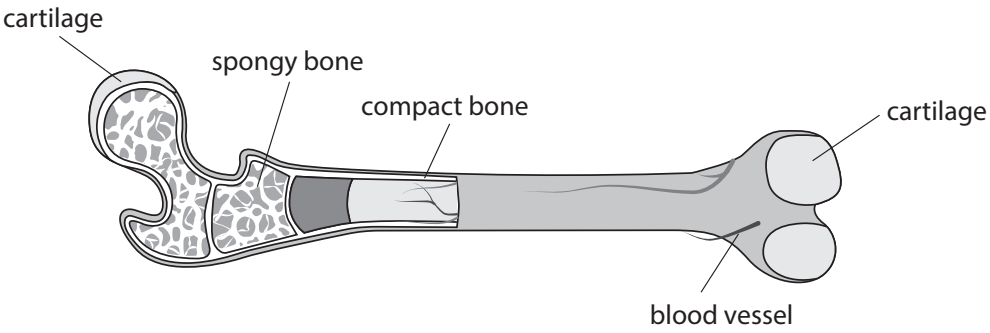
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9. Question 1 · part b.i · 4 marks

June 2025 Paper 2R

Further practice · Attempt part b.i only.

1 (a) The diagram shows a bone found in the human body.



(Source: adapted from: © Designua / Shutterstock)

(i) State the name of the type of bone shown in the diagram.

(1)

.....

(ii) Place one tick (✓) in each row to give the function of each part of the bone.

(3)

Part	Function		
	reduces friction	where red blood cells are produced	provides strength
compact bone			
spongy bone			
cartilage			

(iii) Describe the function of the blood vessel in the bone.

(2)

.....

.....

.....

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9. Question 1 · part b.i · 4 marks

June 2025 Paper 2R

Further practice · Attempt part b.i only.

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(b) Bones are moved by muscles.

(i) Use the words in the box to complete the passage about bones and muscles.

(4)

cartilage	contracts	ligaments	lowers	pulls
pushes	raises	relaxes	tendons	

Muscles are attached to bones by ..... .

To raise the lower arm, the biceps ..... and the triceps ..... .

The biceps ..... on the radius, which raises the lower arm.

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(ii) Energy is transferred by the biceps muscle when raising the lower arm.

The biceps muscle applies a force of 125 N to raise the lower arm a distance of 0.50 m.

Use this formula to calculate the energy transferred by the biceps muscle when raising the lower arm 0.50 m.

energy transferred (J) = force (N) × distance (m)

(2)

energy transferred = ... J

(Total for Question 1 = 12 marks)

9. Question 1 · part b.i · 4 marks · Working and answer

Working and answer

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